
False-Positive-Aware CLIP-YOLO Hybrid Learning for Wildfire Smoke and Fire Detection

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Abstract

Early wildfire monitoring requires detectors that localize both visible flame and diffuse smoke while resisting visually similar false positives such as clouds, haze, fog, glare and waterfalls. This paper studies a YOLOv8n baseline and a proposed CLIP-YOLO hybrid model on the D-Fire smoke/fire detection task. The baseline is first fine-tuned and evaluated as a lightweight real-time detector. We then mine false positives from negative validation images and use them, together with curated negative scenes and language prompts, to construct contrastive anchors from a frozen CLIP encoder. The proposed hybrid injects CLIP-derived semantic context into the YOLO P5 feature map using a fusion head and trains with a joint detection and contrastive objective. Experiments show that YOLOv8n remains the strongest model, reaching $mAP@0.5 = 0.7057$ and $F1 = 0.6718$. The selected best CLIP-YOLO hybrid checkpoint from the epoch sweep reaches $mAP@0.5 = 0.4914$ and $F1 = 0.5066$ at epoch 12, while longer hybrid training showed performance falloff. These results indicate that false-positive-aware semantic supervision is promising but sensitive: it can improve some hybrid configurations over weaker settings, yet it can also disturb class balance and reduce smoke recall. The paper reports the full architecture, training objective, ablations, qualitative outputs and failure modes.

1. Introduction and Motivation

Wildfire smoke and fire detection is a practical object-detection problem with strong safety relevance. Early response depends not only on identifying intense flames, but also on localizing low-contrast smoke plumes before the event becomes visually obvious. In surveillance or remote-camera settings, the visual signal is ambiguous: clouds, fog, haze, sun glare, waterfalls, dust, distant hills and urban atmospheric effects can resemble smoke. A detector that performs well on clean flame images may still raise false alarms or miss faint smoke in realistic scenes.

Our central motivation was therefore not simply to train a higher-capacity detector, but to ask whether semantic supervision from a vision-language model could help an object detector become less vulnerable to smoke-like negatives. We began with a YOLOv8n detector because it is small, fast and suitable for real-time deployment. After evaluating the baseline, we mined examples where the trained detector produced false positives on negative images. These mined errors were then used to form negative CLIP anchors, with the goal of explicitly pushing the hybrid model away from common smoke-like distractors.

The final result is intentionally reported honestly: the CLIP-YOLO hybrid was successfully implemented and optimized, but did not outperform the YOLOv8n baseline. This negative result is useful because it reveals a tension between detection localization and global semantic alignment. CLIP features can provide useful high-level context, but if fused too aggressively or trained with poorly balanced anchors, they can weaken smoke localization and class balance.

2. Related Work

One-stage object detectors such as YOLO are widely used when speed and localization must be balanced. YOLOv8, as implemented by Ultralytics, uses a modern real-time detection pipeline with a lightweight nano variant suitable for constrained environments. In this work, YOLOv8n serves as the detector backbone and baseline.

CLIP introduced large-scale contrastive language-image pre-training, learning a shared embedding space for images and text. Its zero-shot and semantic transfer abilities make it attractive for tasks where labeled visual examples are limited or where semantic categories need additional language grounding. However, CLIP is not itself a dense localizer; using it inside an object detector requires architectural care. Our design therefore keeps CLIP frozen and uses it as a semantic anchor rather than replacing the YOLO detection head.

Fire and smoke detection datasets are challenging because they include both object-level annotations and negative scenes. The D-Fire dataset provides images and bounding boxes for smoke and fire, making it suitable for evaluation.

ing object detectors rather than only image-level classifiers. Our work uses this dataset to evaluate both the baseline and hybrid variants.

3. Dataset and Experimental Setup

We used the D-Fire smoke/fire dataset in YOLO format with two target classes: smoke and fire. The dataset split used in the notebook contained 14,122 training images, 3,099 validation images and 4,306 test images. The training set contained 7,664 labeled fire/smoke images and 6,458 background images. During evaluation, 15 test images were ignored by the Ultralytics loader because of corrupt labels or out-of-bounds normalized coordinates, leaving 4,295 scanned test images.

Images were resized to 640×640 . Baseline evaluation used batch size 16, while hybrid training used batch size 8 due to the additional CLIP preprocessing and fusion path. The experiments ran in Kaggle on a Tesla T4 GPU with PyTorch and Ultralytics YOLO. The baseline weights were stored as best.pt and reused to initialize the hybrid model.

3.1. Main quantitative results

Model	Epoch	λ	α	mAP50	mAP50-95	Prec.	Rec.	F1	FPS
YOLOv8n Baseline	baseline	–	–	0.7057	0.3912	0.7112	0.6365	0.6718	59.1285
CLIP+YOLOv8n Hybrid	12	0.5	0.5	0.4914	0.2194	0.5338	0.4821	0.5066	81.7615
Delta Hybrid - Baseline	–	–	–	-0.2143	-0.1718	-0.1774	-0.1544	-0.1651	22.6330

Table 1. Main quantitative comparison from the uploaded baseline-vs-best-hybrid CSV. The selected hybrid checkpoint is epoch 12 with $\lambda = 0.5$ and $\alpha = 0.5$; the detector-only YOLOv8n baseline remains strongest overall.

4. Methodology

4.1. Baseline detector

The baseline uses YOLOv8n, selected for its real-time detection behavior, small parameter count and suitability for edge-style fire monitoring. The detector is fine-tuned on D-Fire and evaluated on the held-out test split. It predicts bounding boxes and class scores for smoke and fire using the standard YOLO detection objective.

4.2. False-positive mining

After training the baseline, we ran it on negative validation images using a confidence threshold of 0.25. If a validation image had an empty label file but the detector produced at least one prediction, we stored that image path, number of detections and maximum confidence. This produced 35 mined false-positive examples in the saved manifest. The test split was not used for mining, avoiding evaluation leakage.

Mined False Positives — what fools the trained baseline

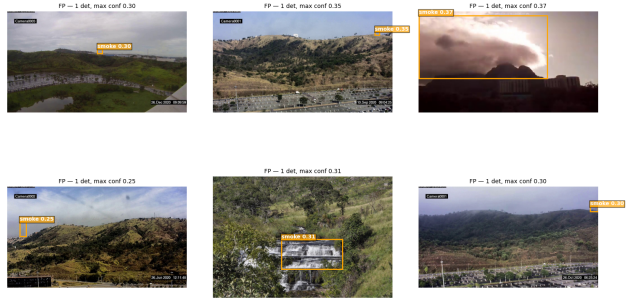


Figure 1. Mined false positives: cloud, haze, glare, waterfall and atmospheric patterns that fooled the trained baseline into predicting smoke. These examples motivated false-positive-aware negative anchors.

4.3. CLIP-guided negative and positive anchors

We used a frozen CLIP ViT-B/32 encoder. Positive fire/smoke anchors were built from 16 fire-related prompts. Negative anchors were built from 16 failure-mode-aware negative prompts and image embeddings from mined false positives and curated negative scenes. The hyperparameter alpha controls the contribution of image-derived negatives versus text-derived negatives. Alpha = 0 is text-only, alpha = 1 is image-only, and alpha = 0.5 is a balanced mixture.

This design was chosen because the baseline errors were not random. They were semantically meaningful: smoke-like textures and atmospheric effects. A text/image contrastive anchor allows the training objective to express that such patterns should not be treated as smoke, even if they share low-level visual texture with smoke.

4.4. Proposed CLIP-YOLO architecture

The hybrid model has two parallel branches. The left branch is the YOLOv8n backbone and neck, which produces multi-scale feature maps P3, P4 and P5. P3, P4 and P5 feed the detection head. The right branch is frozen CLIP, which encodes the input image into a semantic vector. The CLIP vector is projected to the YOLO channel dimension and concatenated with the YOLO P5 feature map. A 1×1 convolutional fusion head produces a fused P5 map, which is then used by the YOLO detection head and by a contrastive projection head.

We fused at P5 because P5 contains the highest-level YOLO representation and is therefore the most compatible location for global CLIP context. Lower feature maps such as P3 and P4 preserve fine spatial detail, so altering them directly could disrupt small-object localization. P5 fusion lets the detector keep multi-scale localization while giving the high-level representation access to semantic cues.

Proposed CLIP-YOLO Hybrid Architecture and Training Objective

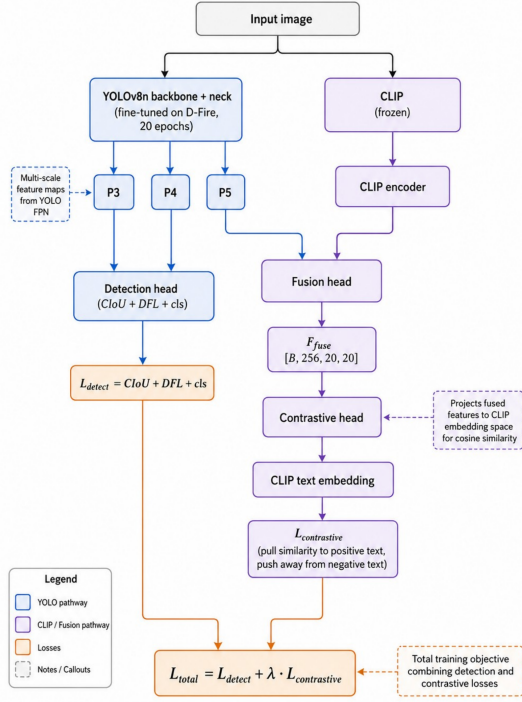


Figure 2. Proposed CLIP-YOLO hybrid architecture and training objective.

4.5. Training objective

The hybrid objective combines the normal detection loss and a contrastive loss: $L_{total} = L_{detect} + \lambda L_{contrastive}$. The detection loss is the YOLO loss consisting of Complete Intersection over Union (CIoU), Distribution Focal Loss (DFL) and classification loss. The contrastive loss projects fused features into the CLIP embedding space, pulling images with fire/smoke labels toward the positive anchor and pulling negative/background images toward the negative anchor.

Hybrid training used a two-phase strategy. During Phase A, YOLO parameters were frozen and only the new fusion and contrastive heads were trained. During Phase B, YOLO was unfrozen for joint optimization. The selected best hybrid checkpoint used $\lambda = 0.5$ and $\alpha = 0.5$ at epoch 12 in the epoch sweep. The longer 25-epoch run used $\lambda = 0.5$, $\alpha = 0.5$ and 5 frozen epochs to test whether extended joint optimization could help, but it showed performance falloff. The ablation runs used 10 epochs with 3 frozen epochs.

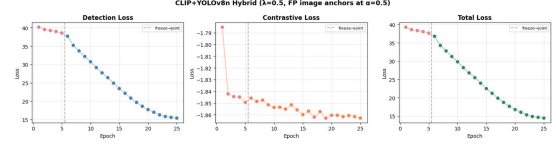


Figure 3. Training curves for the 25-epoch CLIP-YOLO hybrid. Detection and total loss decrease after the freeze-to-joint transition, showing that the model optimized successfully even though final detection performance degraded.

5. Implementation Snippets

The following snippets show the core implementation ideas. They are shortened from the notebooks to preserve the conference-style flow while retaining reproducibility of the architecture and objective.

5.1. Fusion and contrastive projection heads

```
class FusionHead(nn.Module):
    def __init__(self, yolo_ch=256, clip_dim=512):
        super().__init__()
        self.clip_proj = nn.Linear(clip_dim, yolo_ch)
        self.fuse_conv = nn.Conv2d(yolo_ch * 2, yolo_ch,
                                   kernel_size=1, bias=False)
        self.bn = nn.BatchNorm2d(yolo_ch)
        self.act = nn.SiLU()
        # identity initialization: preserve YOLO P5 at epoch 0
        with torch.no_grad():
            self.fuse_conv.weight.zero_()
            for c in range(yolo_ch):
                self.fuse_conv.weight[c, c, 0, 0] = 1.0
    def forward(self, yolo_feat, clip_vec):
        B, C, H, W = yolo_feat.shape
        clip_proj = self.clip_proj(clip_vec).view(B, C, 1, 1).
        expand(B, C, H, W)
        fused = torch.cat([yolo_feat, clip_proj], dim=1)
        return self.act(self.bn(self.fuse_conv(fused)))

class ContrastiveHead(nn.Module):
    def __init__(self, yolo_ch=256, clip_dim=512):
        super().__init__()
        self.proj = nn.Linear(yolo_ch, clip_dim)
    def forward(self, fused_feat):
        pooled = fused_feat.mean(dim=[2, 3])
        return F.normalize(self.proj(pooled), dim=-1)
```

5.2. Contrastive loss

```
def contrastive_loss(yolo_embed, has_fire, text_fire, text_neg,
                    temperature=1.0):
    text_fire = text_fire.to(yolo_embed.dtype)
    text_neg = text_neg.to(yolo_embed.dtype)
    fire_sim = (yolo_embed @ text_fire.T).squeeze(-1)
    neg_sim = (yolo_embed @ text_neg.T).squeeze(-1)
    pos_mask = has_fire.bool()
    neg_mask = ~pos_mask
    loss = torch.tensor(0.0, device=yolo_embed.device, dtype=
        yolo_embed.dtype)
    if pos_mask.sum() > 0:
        loss = loss - fire_sim[pos_mask].mean()
    if neg_mask.sum() > 0:
        loss = loss - neg_sim[neg_mask].mean()
    return loss / temperature
```

5.3. Joint training objective

```
for epoch in range(NUM_EPOCHS):
    if epoch == FREEZE_EPOCHS:
        for name, param in hybrid_model.named_parameters():
            if name.startswith('yolo.'):
                param.requires_grad = True
        preds, yolo_embed = hybrid_model(yolo_imgs, clip_imgs)
        L_detect, _ = detect_loss_fn(preds, batch_dict)
```

```

L_contrast = contrastive_loss(yolo_embed, has_fire, TEXT_FIRE
, TEXT_NEG)
L_total = L_detect + LAMBDA_1 * L_contrast
optimizer.zero_grad()
L_total.backward()
torch.nn.utils.clip_grad_norm_(hybrid_model.parameters(),
max_norm=10.0)
optimizer.step()

```

6. Results

6.1. Baseline performance

The YOLOv8n baseline achieved the best overall performance. It reached $\text{mAP}@0.5 = 0.7057$, $\text{mAP}@0.5:0.95 = 0.3912$, precision = 0.7112, recall = 0.6365 and $F1 = 0.6718$. Class-wise AP was stronger for smoke (0.7624) than fire (0.6490). This indicates that the baseline learned broad smoke regions well while still handling many flame instances.

The baseline confusion matrix supports this observation. It correctly predicted 1,709 smoke instances and 1,760 fire instances. However, it also missed 583 smoke instances and 1,090 fire instances as background, while background was falsely predicted as smoke 629 times and as fire 846 times. This confirms that the model is useful but not failure-free, especially around background confusion and missed fire cases.

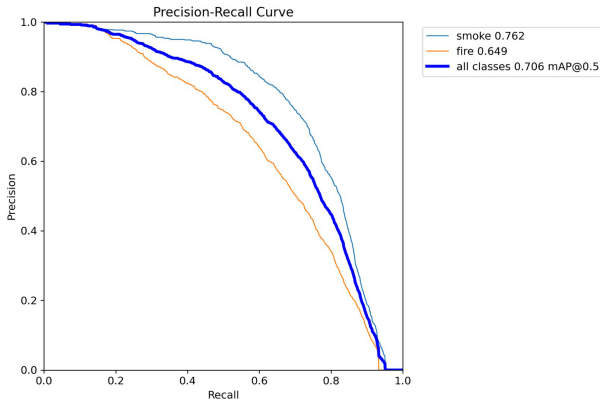


Figure 4. Baseline precision-recall curve. The all-class $\text{mAP}@0.5$ is 0.706, with smoke AP = 0.762 and fire AP = 0.649.

6.2. Hybrid performance

The selected best CLIP-YOLO hybrid checkpoint from the uploaded baseline-vs-best-hybrid comparison used $\lambda = 0.5$, $\alpha = 0.5$ and epoch 12. It reached $\text{mAP}@0.5 = 0.4914$, $\text{mAP}@0.5:0.95 = 0.2194$, precision = 0.5338, recall = 0.4821, $F1 = 0.5066$ and FPS = 81.7615. This did not outperform the baseline, which remained stronger across the main detection metrics. The 25-epoch hybrid run was useful diagnostically, but its later performance falloff suggests overfitting or drift during extended joint optimization.

The ablation sweeps still provide useful evidence about the CLIP-fusion design, but the selected best hybrid checkpoint for the main quantitative comparison is the epoch-12 $\lambda = 0.5$, $\alpha = 0.5$ model from the uploaded baseline-vs-best-hybrid CSV. In the ablation tables, some settings appear stronger under their specific sweep conditions, yet the overall conclusion remains unchanged: the hybrid did not surpass the YOLOv8n baseline.

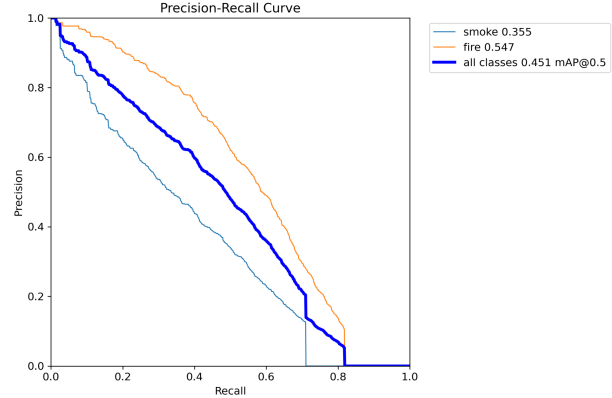


Figure 5. Precision-recall curve for the selected epoch-12 CLIP-YOLO hybrid checkpoint.

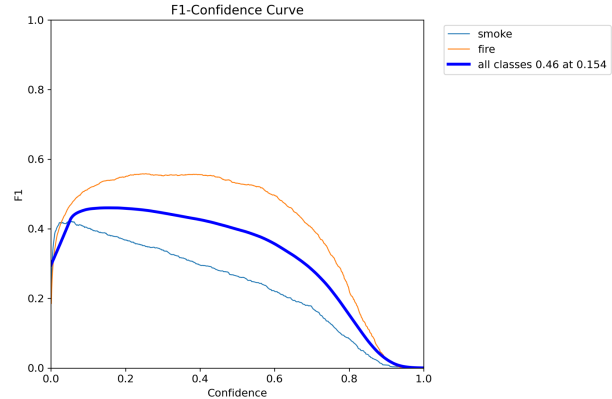


Figure 6. F1-confidence curve for the selected epoch-12 CLIP-YOLO hybrid checkpoint.

7. Ablation Study

We organized the ablation experiments into three tracks. The first track, only `_fp_mining`, used only false positives mined from the baseline detector. These mined false positives were added to the non-fire / non-smoke negative CLIP vectors, and this version did not use hard-negative mining. The second track, `lambda_ablation_fpmining_negmining`, is the final ablation setup: it used both baseline-mined false positives and hard-negative mining while sweeping lambda, the contrastive-loss weight. The third track, `alpha_ablation`, varied alpha, the mixture between text-derived negative an-

chors and image-derived negative anchors. In this setting, alpha controls how much of the negative anchor comes from text prompts versus mined or curated negative image embeddings.

Hard-negative mining was performed manually using hand-picked non-forest fire images from the FASDD dataset, including buildings on fire, cars on fire, urban fires, roadside fires and other non-wildland fire scenes. This choice was deployment-specific. The intended deployment was early forest-fire monitoring in Margalla Hills using PTZ cameras. Because some PTZ cameras may overlook Islamabad or city regions, the model should not trigger forest-fire alarms for urban fires, vehicle fires or building fires in the background. The hard negatives therefore pushed the hybrid toward the target domain: early forest/wildland fire and smoke detection, rather than general fire detection.

7.1. Alpha sweep: text-vs-image negative anchors

α	mAP50	mAP50-95	Prec.	Rec.	F1	AP50 Smoke	AP50 Fire	note
0.0000	0.4910	0.2147	0.5518	0.4952	0.5219	0.4617	0.5202	sweep
0.3000	0.5190	0.2355	0.5751	0.5101	0.5407	0.5071	0.5309	sweep
0.5000	0.5182	0.2392	0.5580	0.5079	0.5318	0.4710	0.5654	main run
0.7000	0.4724	0.2138	0.5154	0.4764	0.4951	0.4287	0.5160	sweep
1.0000	0.4904	0.2238	0.5433	0.4878	0.5141	0.4758	0.5050	sweep

Table 2. Alpha ablation. Alpha = 0.3 gives the best mAP50 and F1 among the alpha sweep results.

The alpha sweep shows that the hybrid is sensitive to the source of the negative anchor. Alpha = 0.3 achieved the best mAP@0.5 and F1, while higher reliance on image-derived false positives degraded performance. This suggests that mined false positives are useful, but if they dominate the negative anchor, they may introduce unstable or over-specific supervision.

7.2. Lambda sweep: only false-positive mining

The only_fp_mining track used baseline-mined false positives only. These false-positive images were added to the non-fire / non-smoke negative CLIP vectors, without the manually curated hard-negative set of urban, vehicle, building, or roadside fires.

λ	mAP50	mAP50-95	Prec.	Rec.	F1	AP50 Smoke	AP50 Fire	note
0.1000	0.4658	0.2082	0.5237	0.4665	0.4934	0.4340	0.4976	sweep
0.3000	0.4881	0.2304	0.5712	0.4739	0.5180	0.5114	0.4647	sweep
0.5000	0.5390	0.2508	0.5795	0.5195	0.5479	0.5049	0.5730	main run
1.0000	0.5269	0.2355	0.5768	0.5243	0.5493	0.5163	0.5374	sweep

Table 3. Lambda ablation with only false-positive mining. This track uses mined baseline false positives but does not use the curated hard-negative set.

In the only-false-positive-mining setup, lambda = 0.5 gives the strongest mAP@0.5, while lambda = 1.0 gives a slightly higher F1. This supports the same overall pattern: contrastive guidance helps most at intermediate strengths and becomes less stable when it dominates training.

7.3. Lambda sweep: FP mining + hard-negative mining

λ	mAP50	mAP50-95	Prec.	Rec.	F1	AP50 Smoke	AP50 Fire	note
0.1000	0.4696	0.2073	0.5608	0.4586	0.5046	0.4314	0.5078	sweep
0.3000	0.4629	0.2088	0.5535	0.4577	0.5011	0.4527	0.4732	sweep
0.5000	0.5367	0.2513	0.5696	0.5354	0.5520	0.5146	0.5588	main run
1.0000	0.4961	0.2249	0.5357	0.5165	0.5260	0.4630	0.5291	sweep

Table 4. Lambda ablation with FP mining and hard-negative mining. Lambda = 0.5 is the strongest setting in this sweep.

The lambda sweep indicates a U-shaped behavior. Very small lambda values underused the CLIP-guided contrastive signal, so the semantic branch did not strongly influence the detector. Very large lambda values disrupted the detection objective and reduced localization/classification performance. Intermediate lambda values gave the best hybrid performance, with lambda = 0.5 reaching mAP@0.5 = 0.5367 in this variant. This suggests that CLIP should act as a regularizer rather than dominate training, and even the strongest hybrid setting still did not surpass the baseline.

8. Qualitative Analysis

Qualitative validation grids show that the baseline and hybrid variants can localize clear fire and smoke regions, but they behave differently. The baseline tends to capture broad smoke plumes better, while hybrid variants often become more fire-sensitive and less reliable on diffuse smoke. Fire predictions are sometimes fragmented into several small boxes, while smoke is sometimes missed or only partially localized.



Figure 7. Example validation labels showing smoke and fire regions in the dataset.



Figure 8. Example predictions from one hybrid/ablation variant. Fire regions are often detected, while smoke localization varies across scenes.

The mined false-positive examples are particularly informative. They show that the trained baseline fires on visual patterns that are not random noise: bright clouds, haze, waterfalls and distant atmospheric regions. This motivated the CLIP-guided contrastive branch, but the final results suggest that preventing such false positives while preserving smoke recall is a harder trade-off than expected.

9. Discussion

The main finding is that the detector-only YOLOv8n baseline is stronger than the proposed CLIP-YOLO hybrid under the tested settings. This is not a failure of implementation: the hybrid trains, its loss decreases and several ablations outperform weaker hybrid settings. Rather, the result shows that semantic alignment and object localization can conflict. CLIP provides image-level semantic information, but fire and smoke detection requires spatial precision, especially for diffuse smoke regions. Global semantic cues may bias the model toward visually salient flames and away from subtle smoke.

The best hybrid results suggest that text prompts are more reliable than image-derived false-positive anchors when used in moderation. $\alpha = 0.3$ performed best in the alpha sweep, meaning that the model benefited from some mined false-positive information but still needed prompt-level semantic structure. This is consistent with the nature of the failure cases: a waterfall or haze image is visually similar to smoke in some local textures, but language can define these as negative concepts more explicitly.

Hard negatives are useful because they encode deployment-specific failure cases, especially the urban and roadside fires that could appear in the background of PTZ camera views. However, these examples also make the learning problem more subtle: the model must distinguish forest-fire-relevant smoke and flame from visually similar but semantically different urban fires. This may require stronger spatial context, scene-level priors, camera metadata, or region-level CLIP supervision rather than only global CLIP embeddings.

For deployment, the baseline is the recommended model because it achieves the highest mAP and F1. The hybrid is best presented as an experimental extension and ablation study. Its contribution is methodological: it demonstrates a false-positive-aware framework that can be improved in future work using better regional CLIP features, prompt ensembling, stronger anchor balancing, or distillation rather than direct feature fusion.

10. Limitations and Future Work

First, the hybrid fuses a global CLIP image vector with a spatial YOLO feature map. This may be too coarse for smoke localization. A region-level or patch-level vision-language model may provide more spatially aligned semantic supervision. Second, the negative anchors are manually designed and mined from a limited set of false positives and hard negatives. A larger and more diverse hard-negative set could improve generalization. Third, the final long hybrid run suggests that more epochs do not automatically improve performance; early stopping and validation-guided checkpoint selection should be added. Finally, threshold tuning should be studied for operational settings where false alarms and missed detections have different costs.

The baseline YOLOv8n detector was trained for 20 epochs before CLIP fine-tuning. This may have made the detector distribution relatively stubborn: once the baseline had already converged, the later CLIP fusion stage may have had limited ability to shift the decision boundary without hurting detection performance. The 25-epoch hybrid experiment was run partly to test whether extended joint optimization could overcome this issue. However, the 25-epoch run appeared to overfit or drift and led to performance falloff.

A better route may be training a fresh YOLOv8n model with CLIP fusion from the beginning, so the detector and CLIP branch co-adapt from early training. This was not done due to compute constraints. Another route is to freeze most or all of YOLO and train only a lightweight CLIP/fusion adapter. Freezing YOLO would preserve the baseline detector’s learned spatial/localization behavior, especially the P3/P4 pathways and detection head. Since CLIP fusion is injected at P5, a frozen-YOLO strategy could reduce drift from the strong baseline while still allowing limited semantic adaptation through the fusion adapter. The tradeoff is that freezing YOLO may preserve baseline accuracy, but it may also limit how much the detector can adapt to CLIP semantics.

11. Conclusion

This paper presented a false-positive-aware CLIP-YOLO hybrid approach for wildfire smoke and fire detection. The baseline YOLOv8n detector achieved strong real-time per-

formance on D-Fire, with $\text{mAP@0.5} = 0.7057$ and $\text{F1} = 0.6718$. The proposed hybrid introduced frozen CLIP features, prompt-based positive/negative anchors, mined false positives and a joint detection-contrastive objective. The selected best hybrid checkpoint achieved $\text{mAP@0.5} = 0.4914$ and $\text{F1} = 0.5066$ at epoch 12, but did not outperform the baseline. The result shows that semantic supervision can be useful, but must be spatially aligned and carefully weighted. The strongest final recommendation is to use YOLOv8n as the deployment model and treat CLIP-guided false-positive mining as a promising future direction rather than a finished improvement.

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